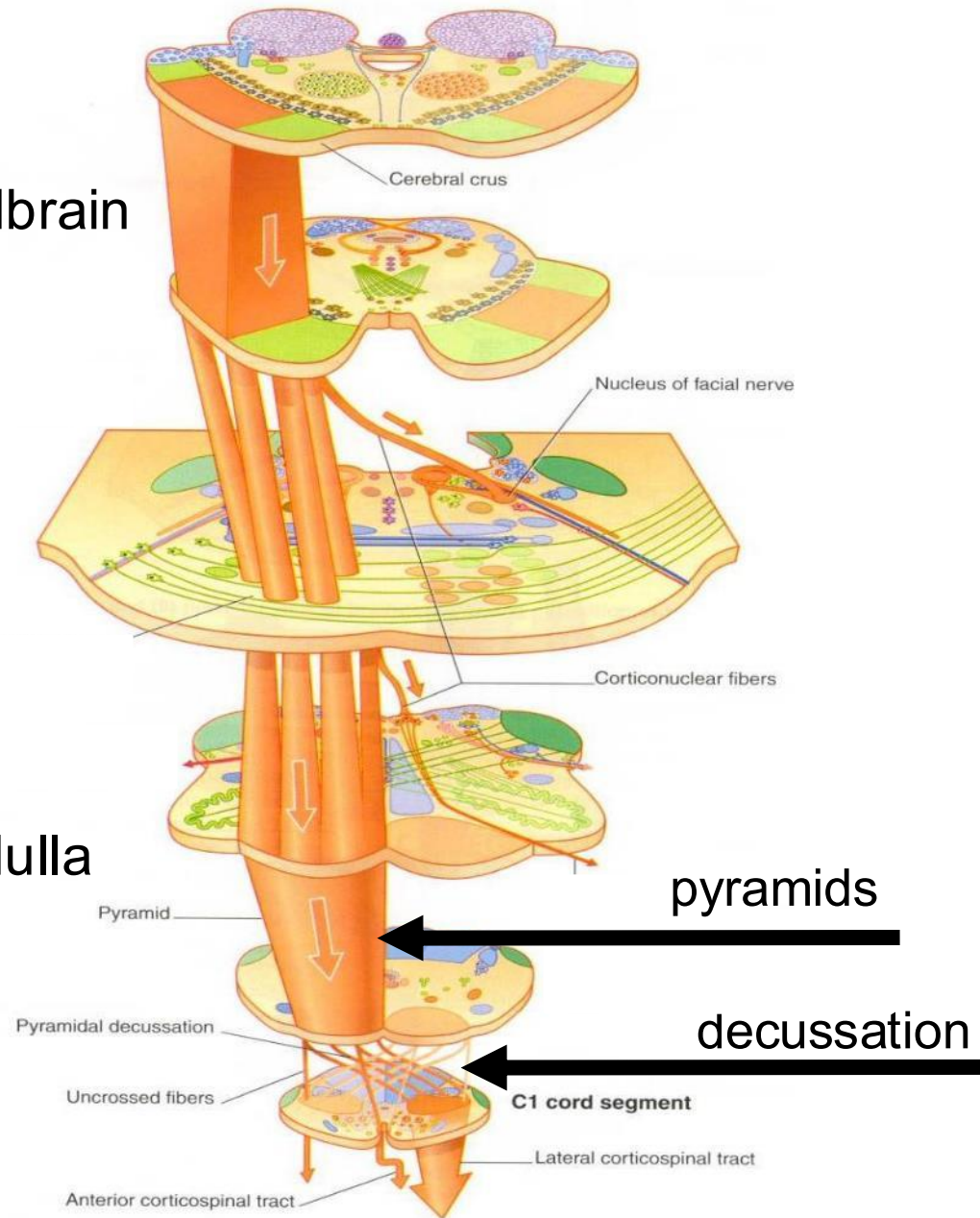


Corticospinal tract

Midbrain

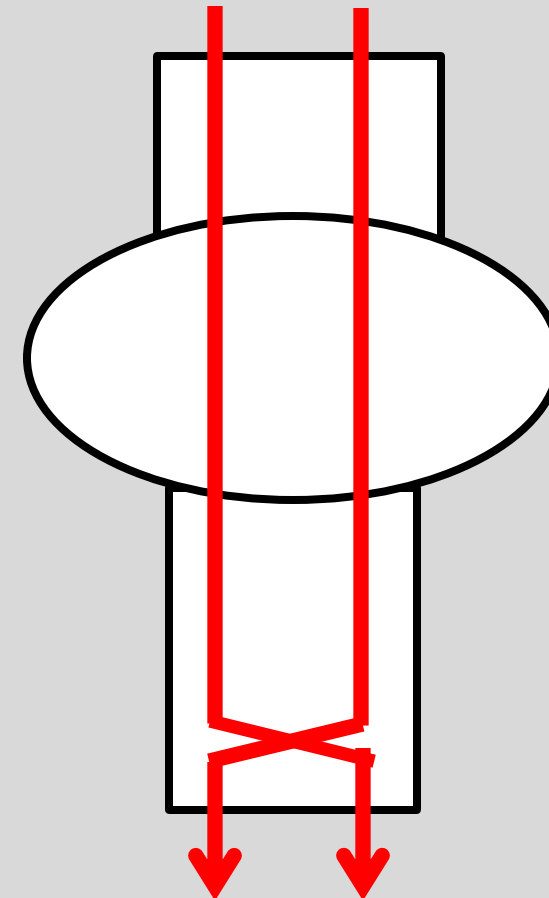
Pons

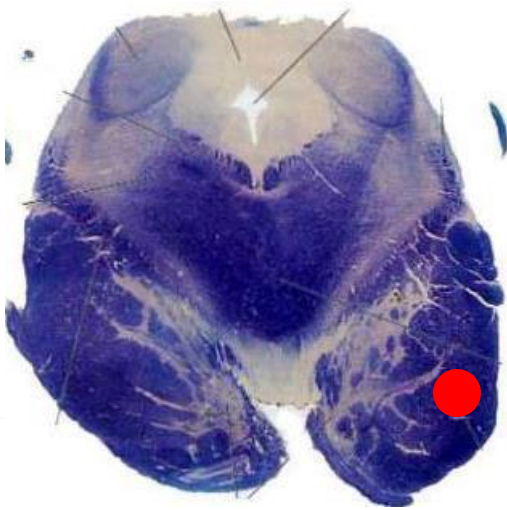
Medulla



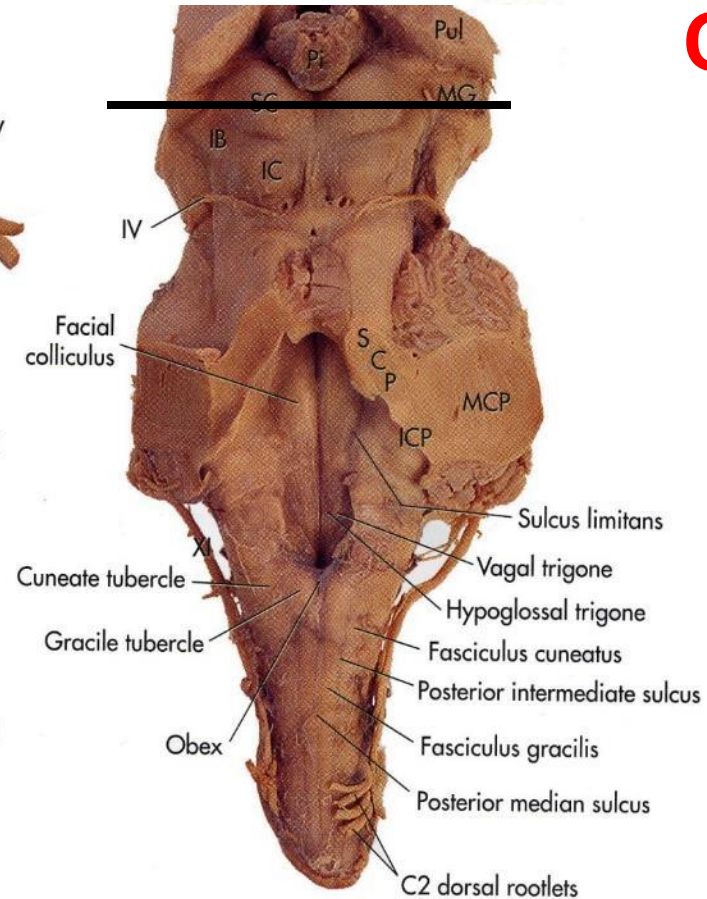
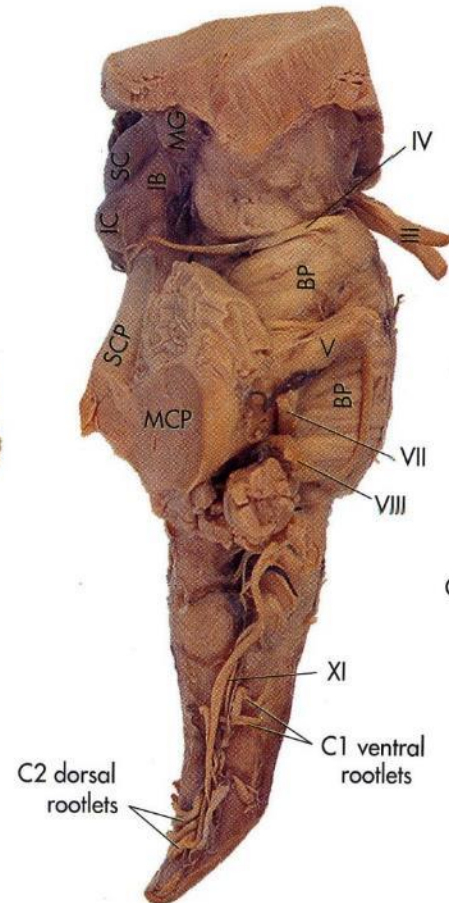
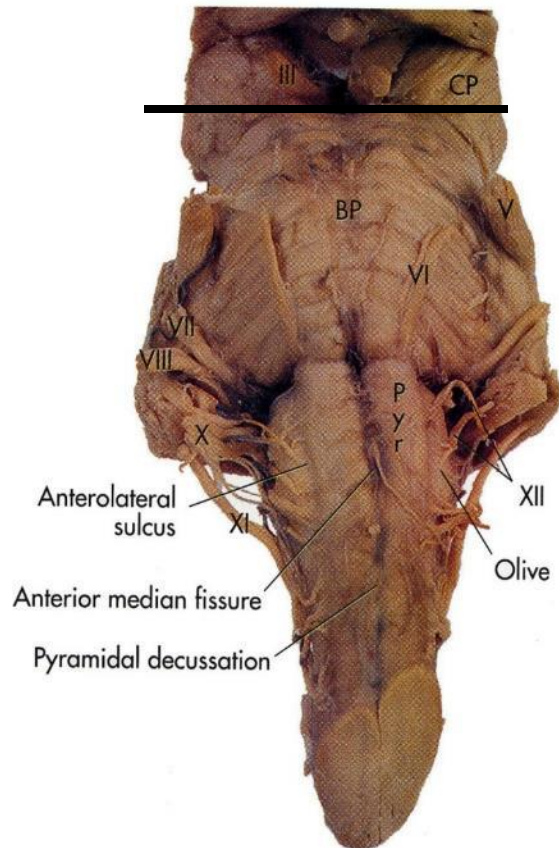
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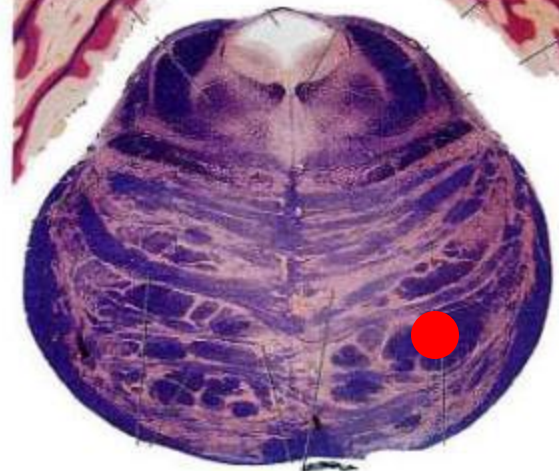
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Corticospinal tract

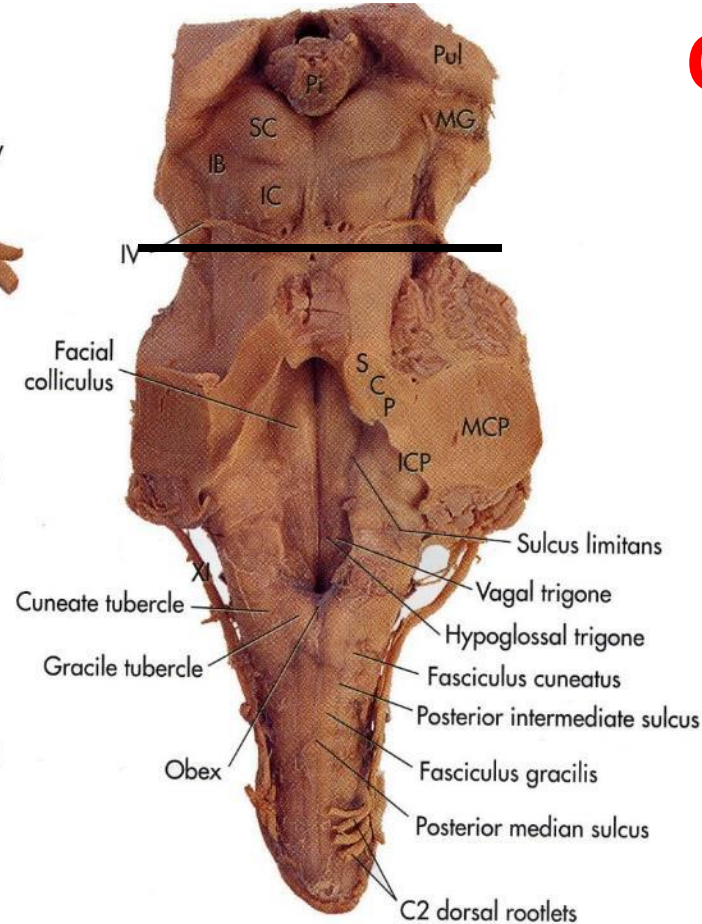
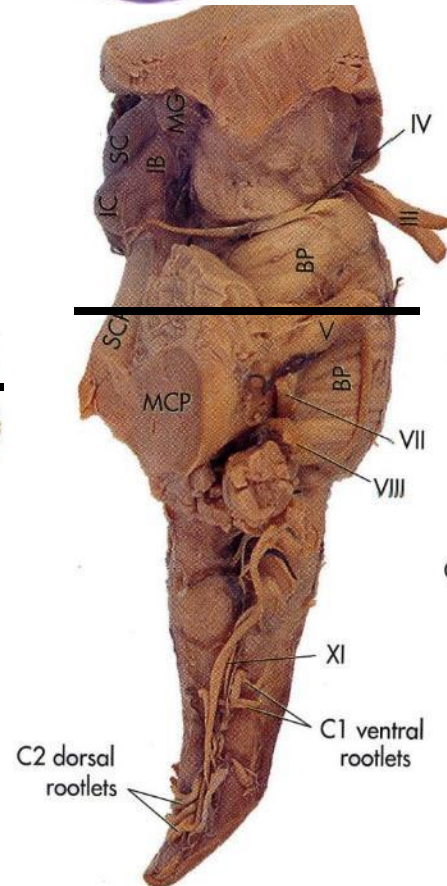
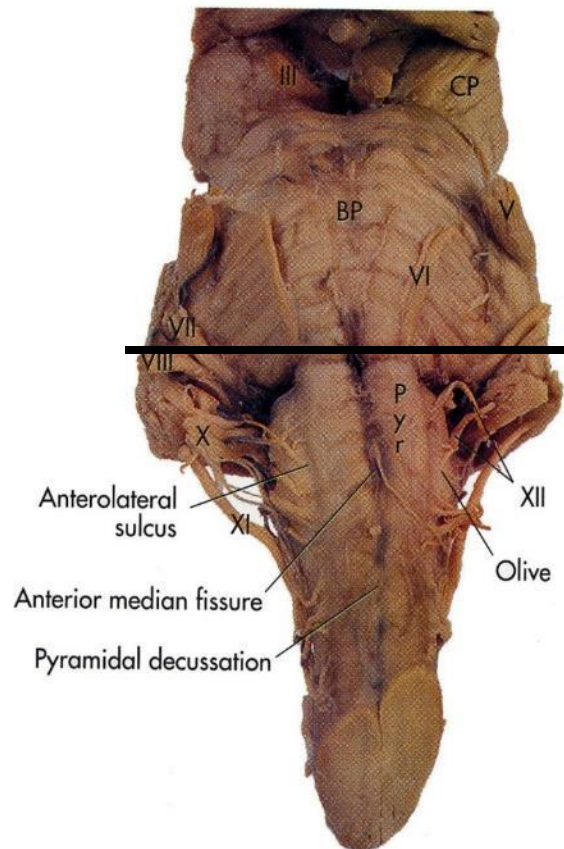


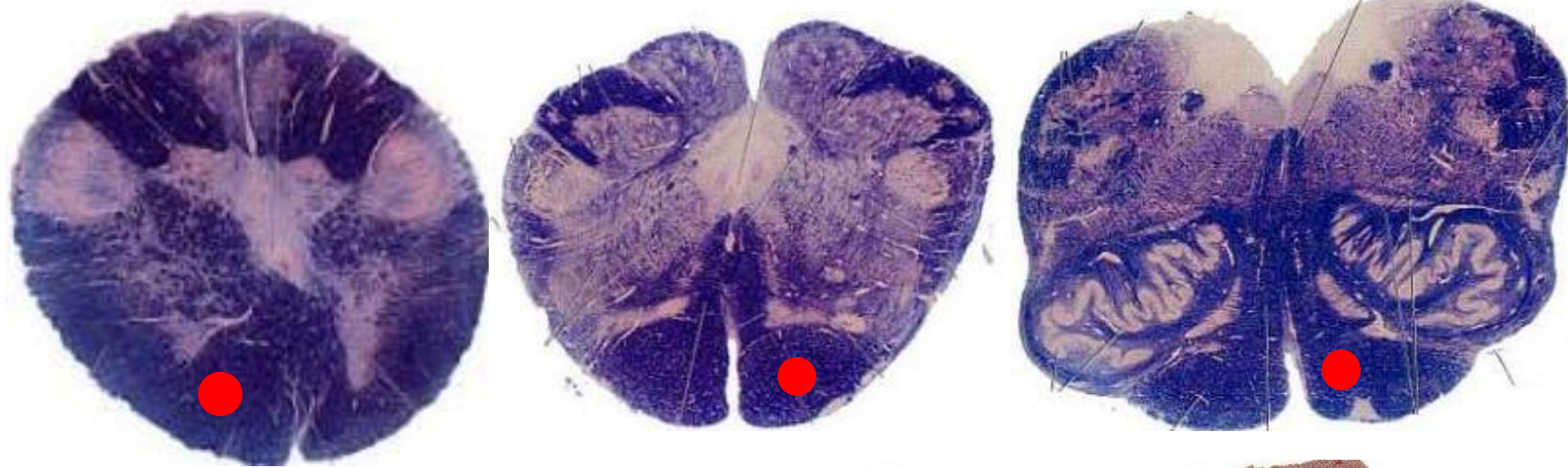


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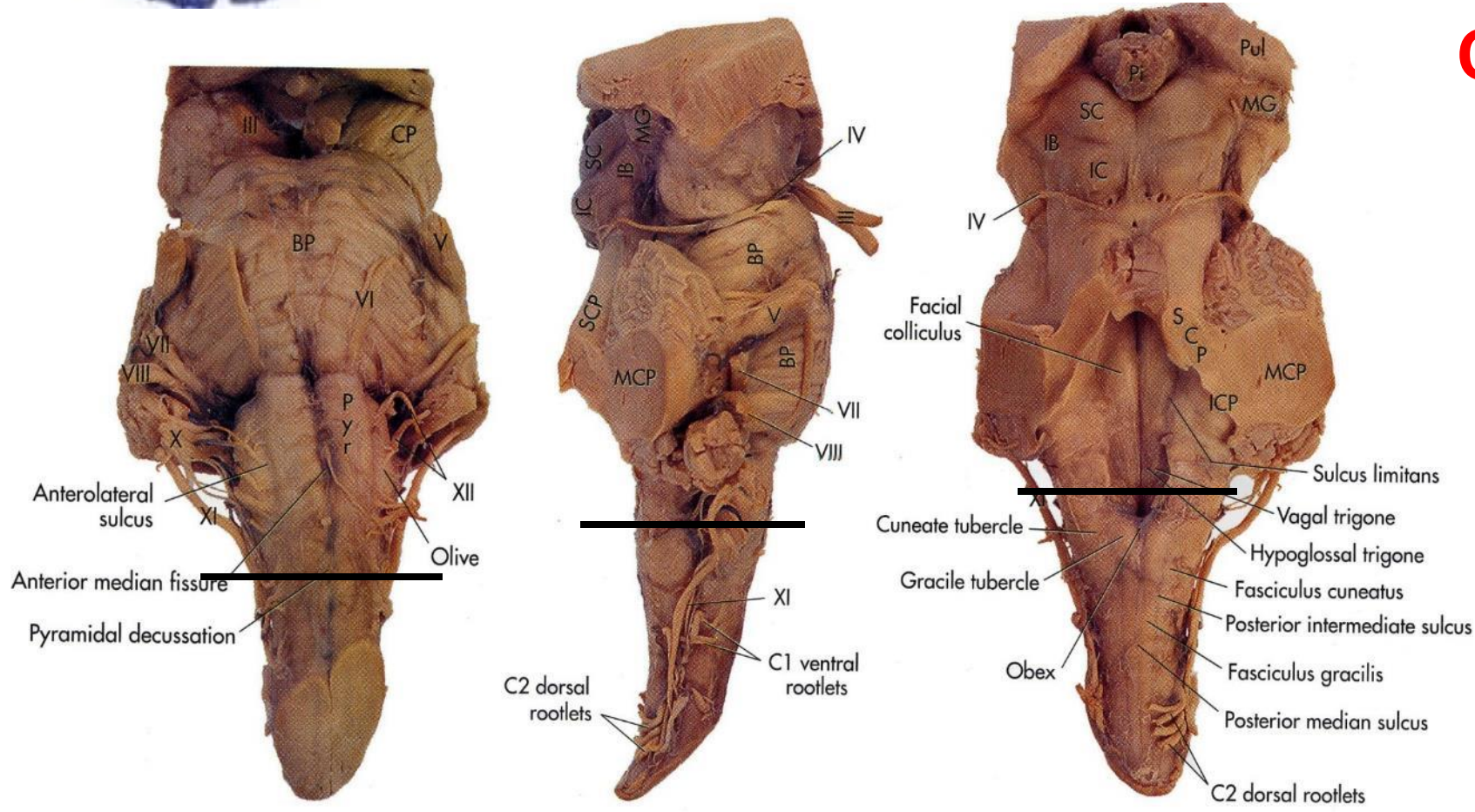




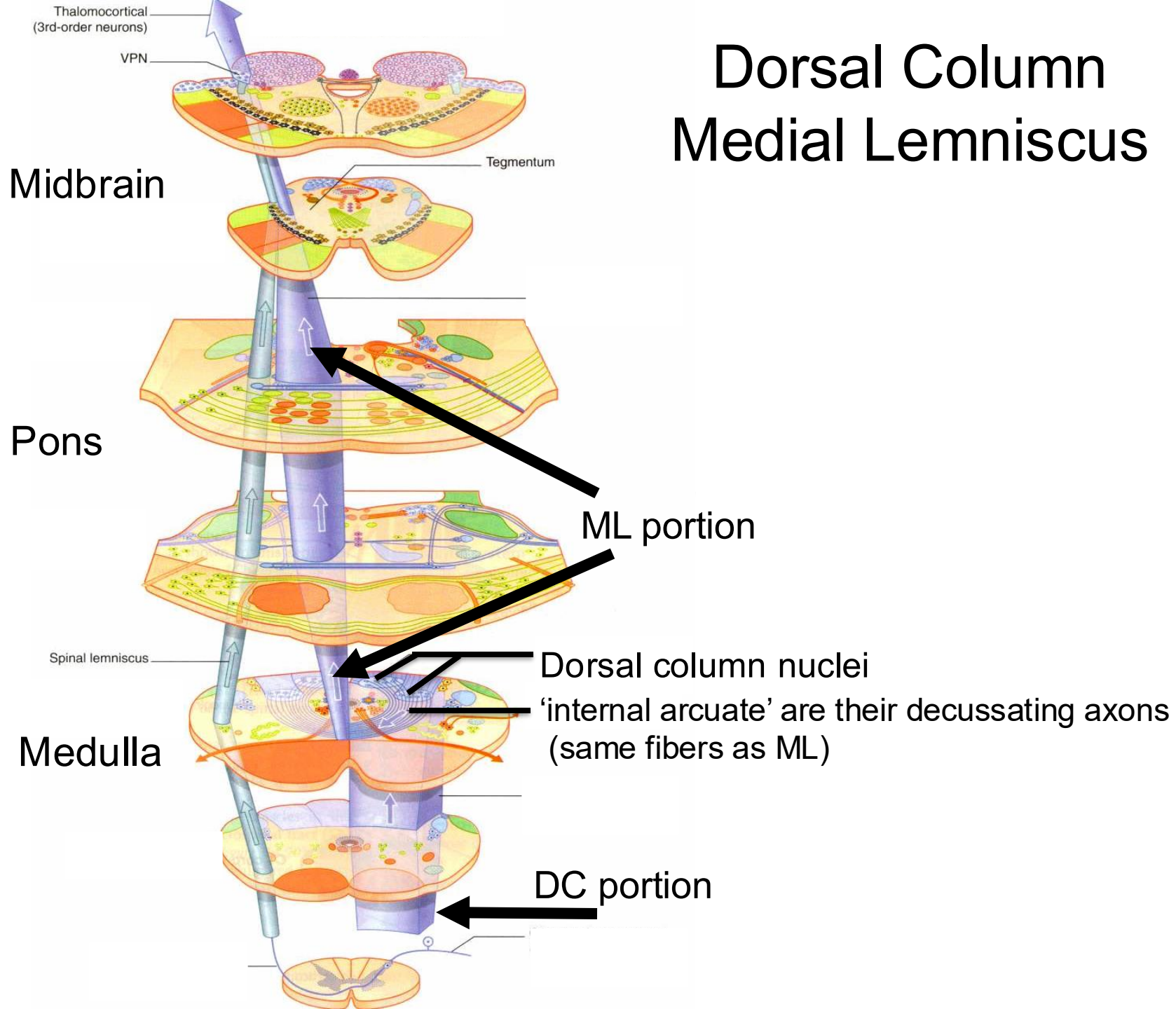
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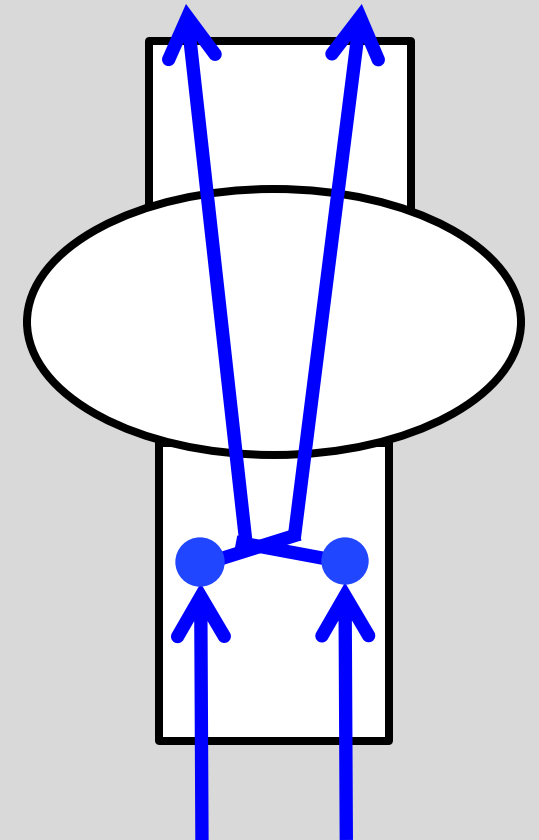


Dorsal Column Medial Lemniscus



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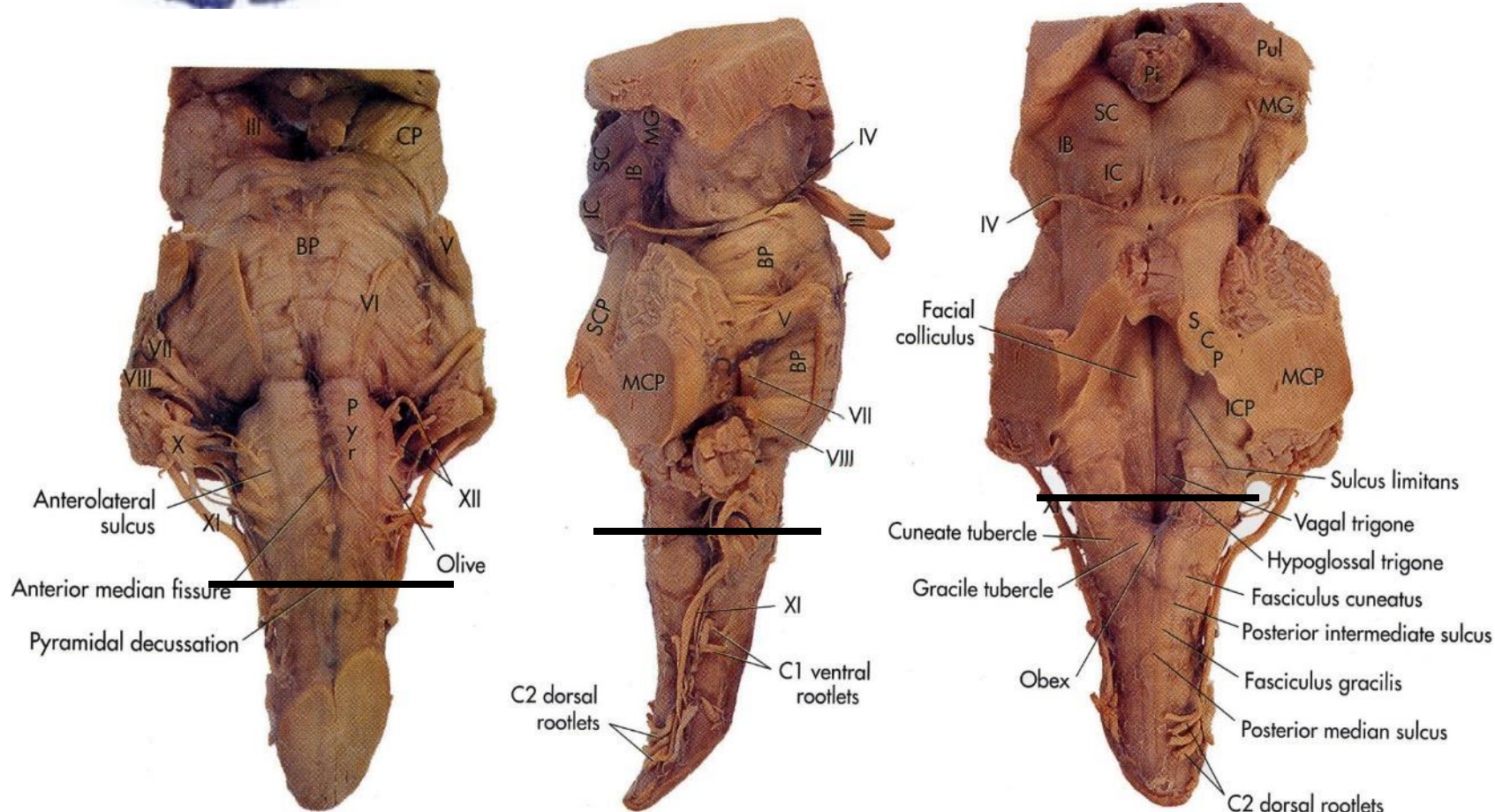
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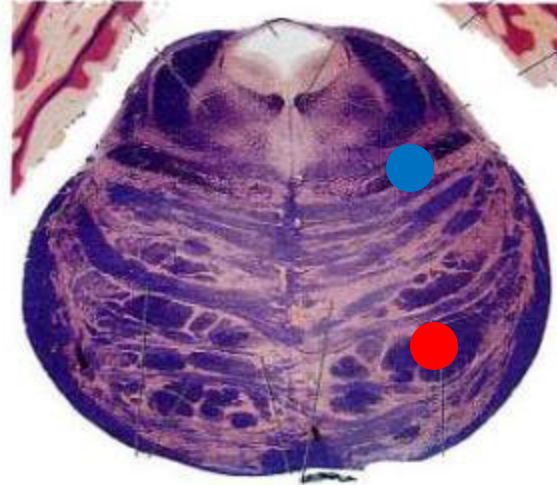
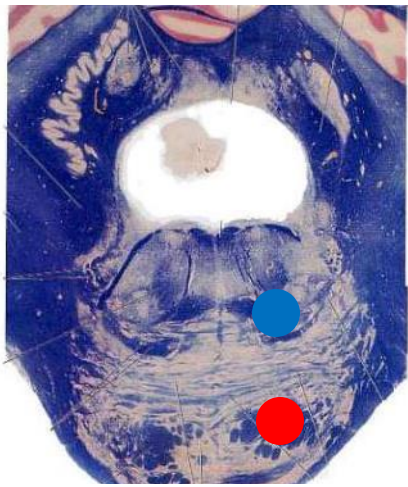


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corticospinal DCML

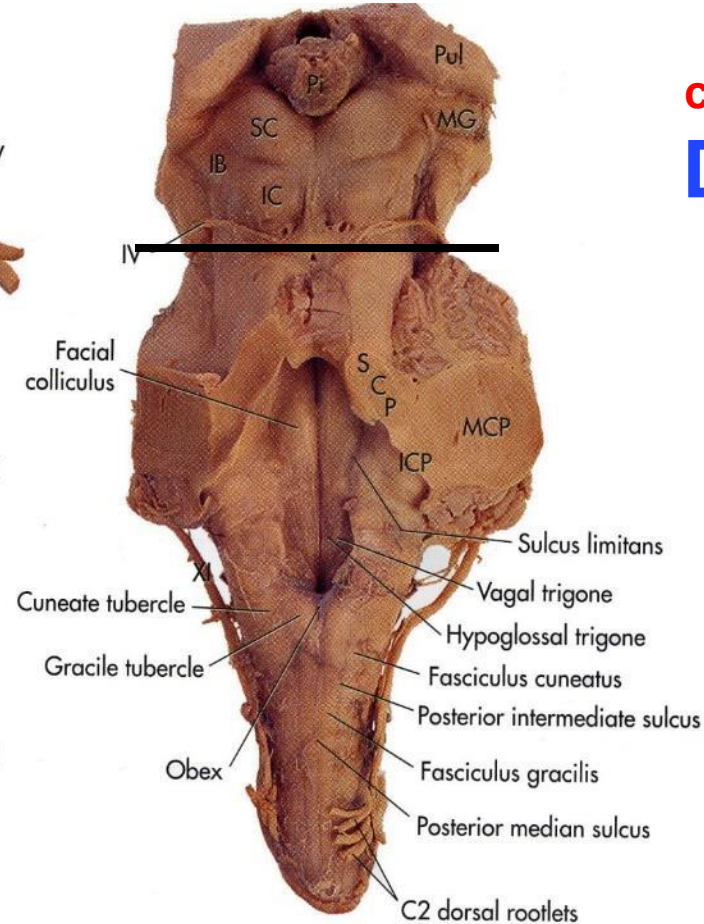
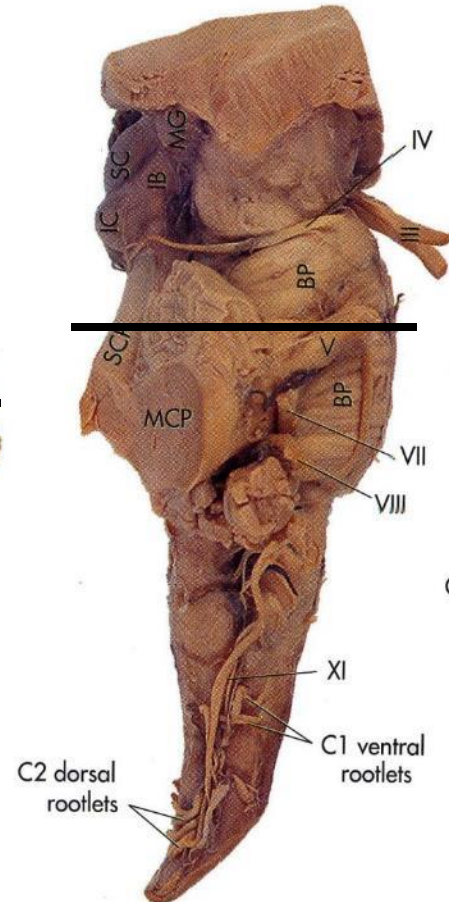
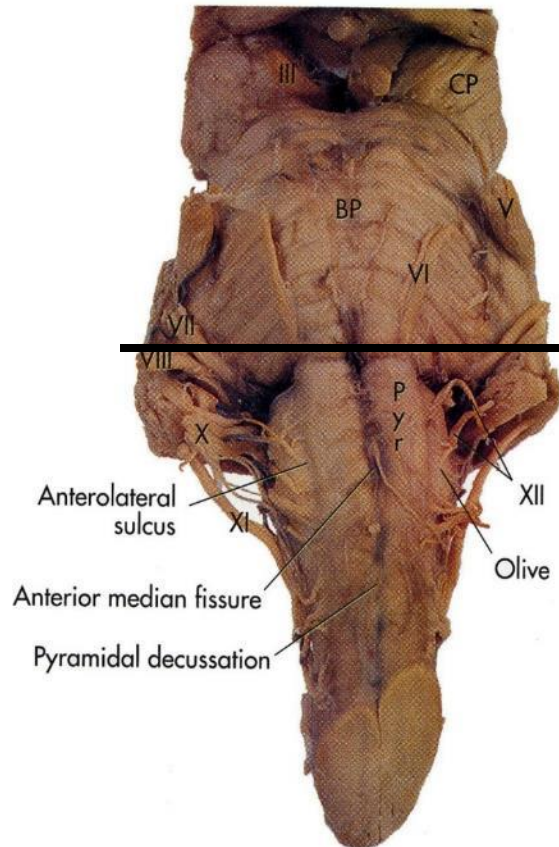


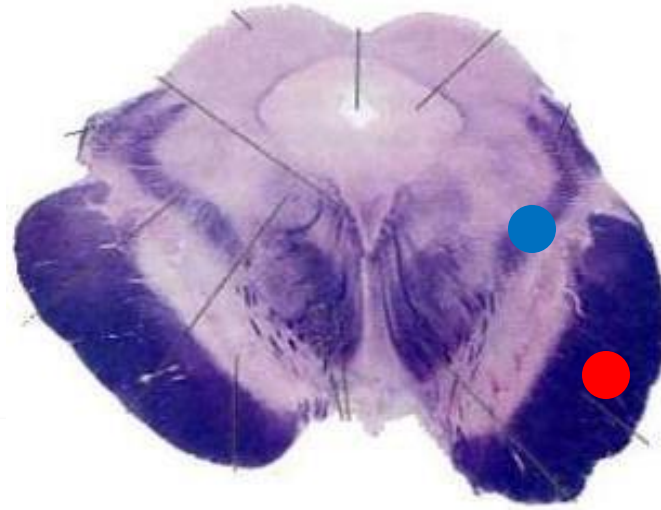
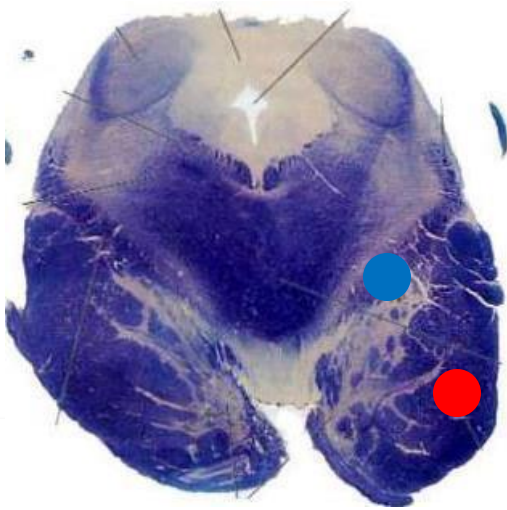


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corticospinal
DCML

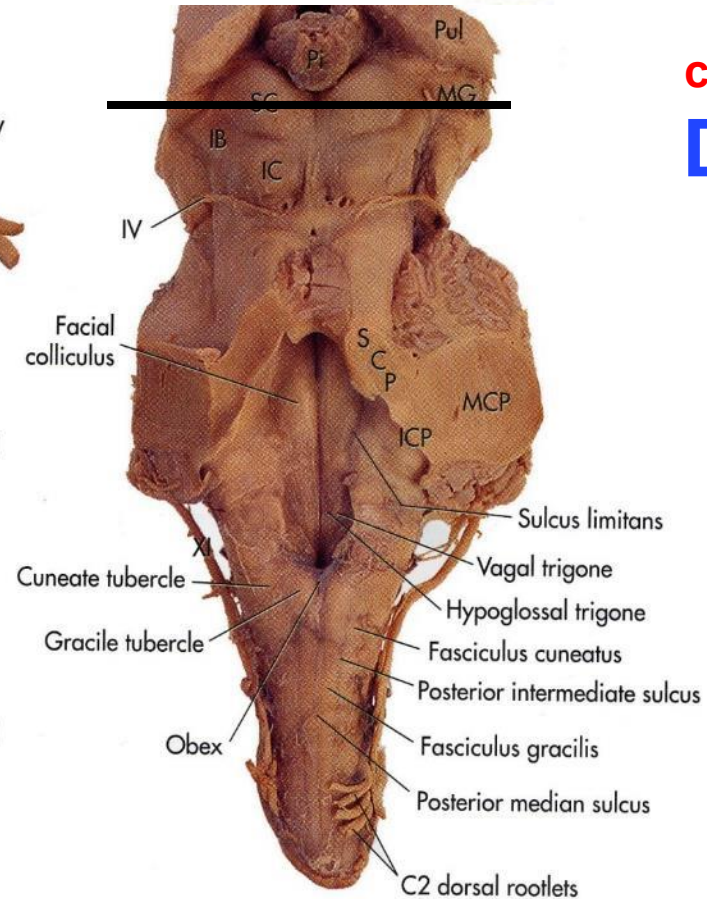
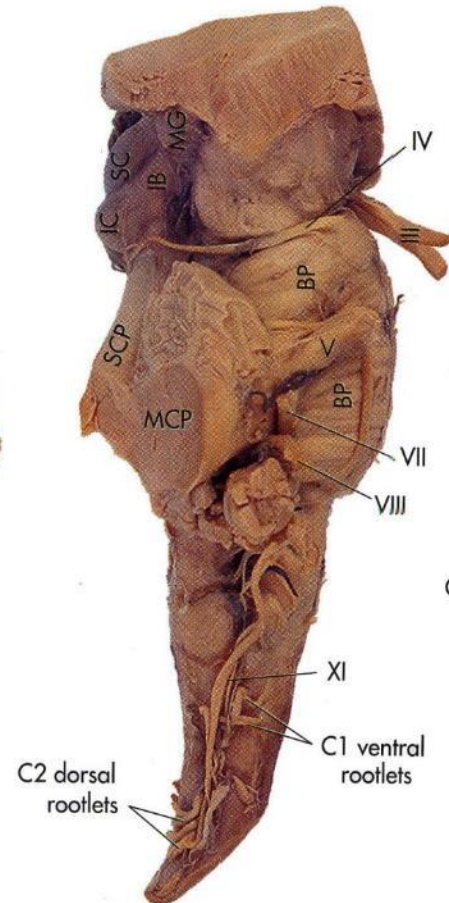
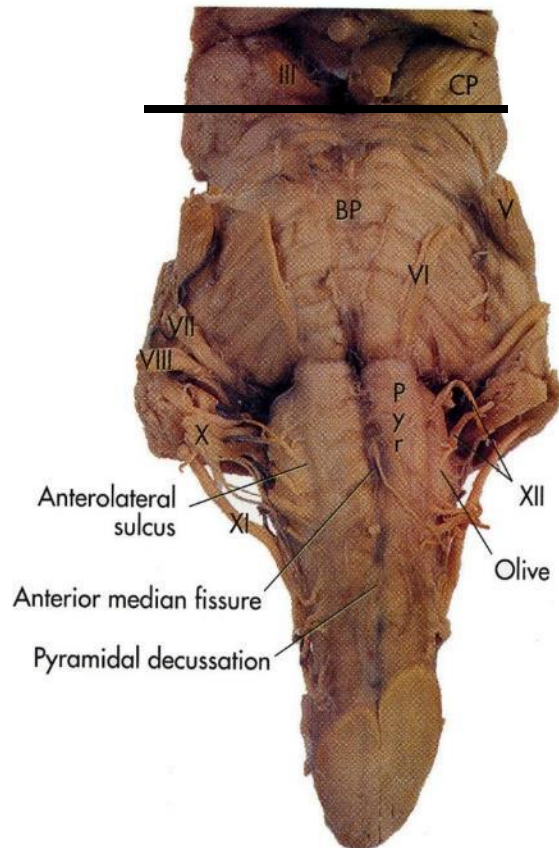




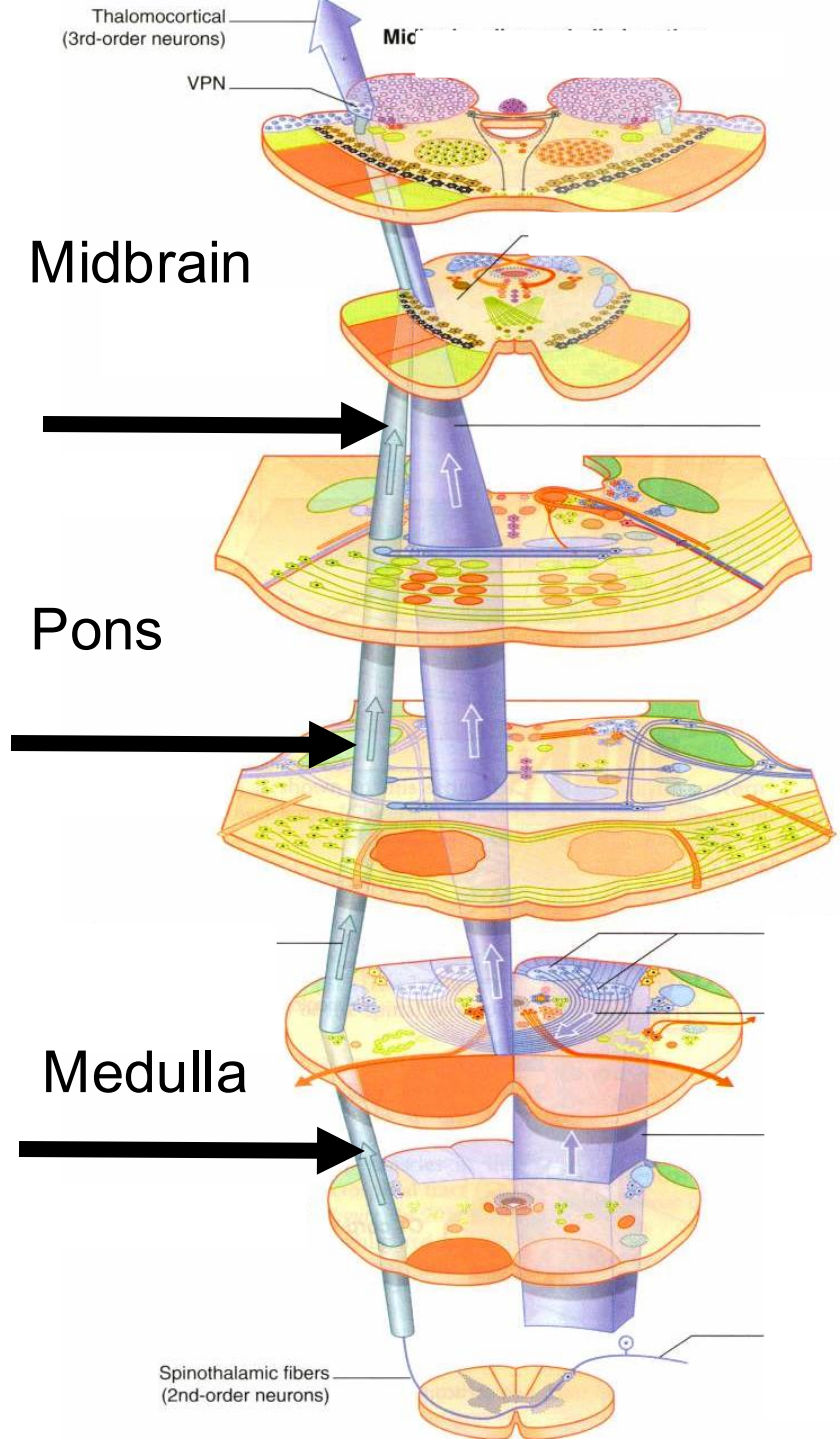
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DCML

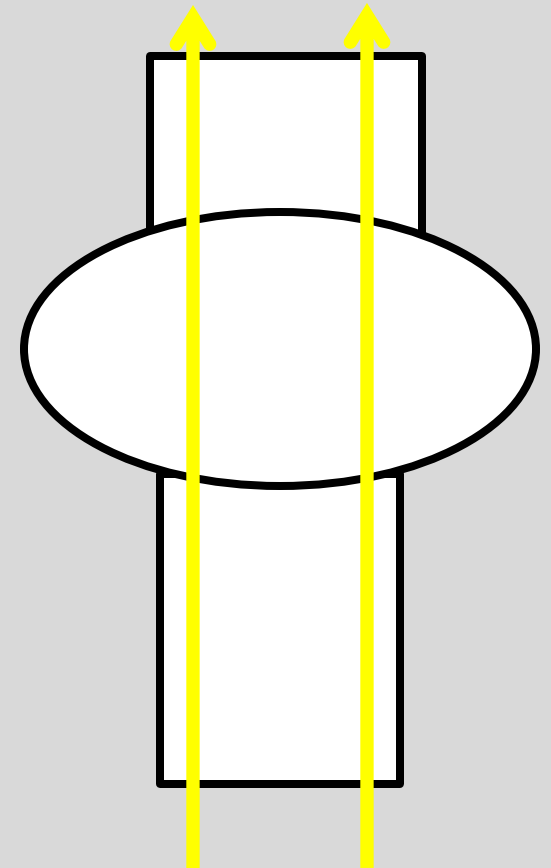


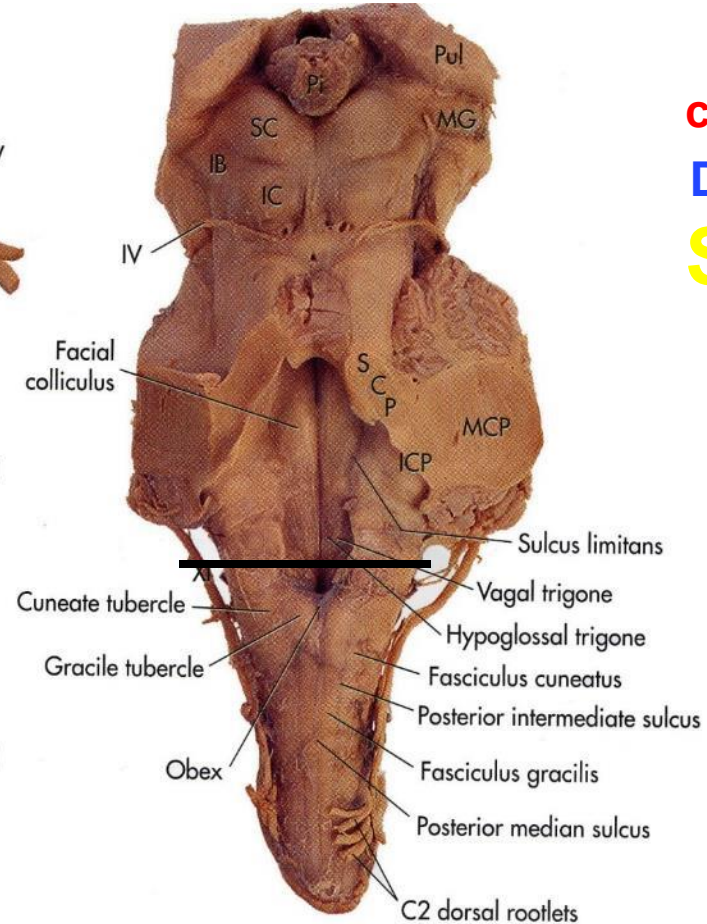
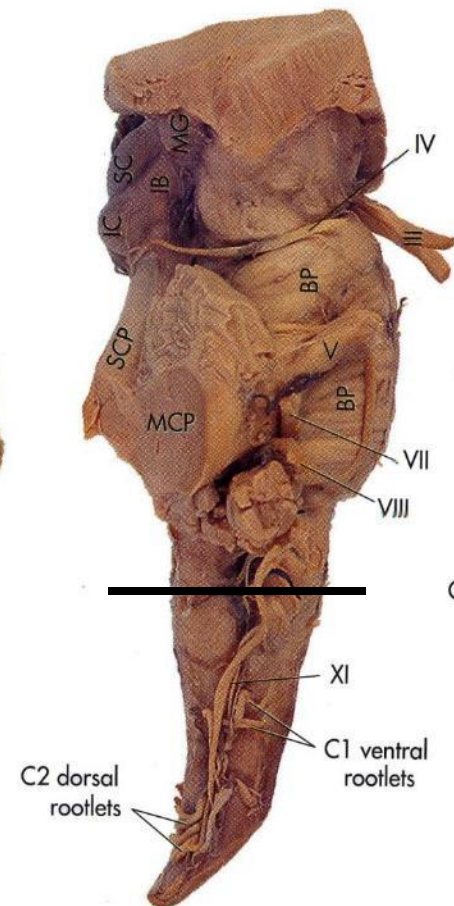
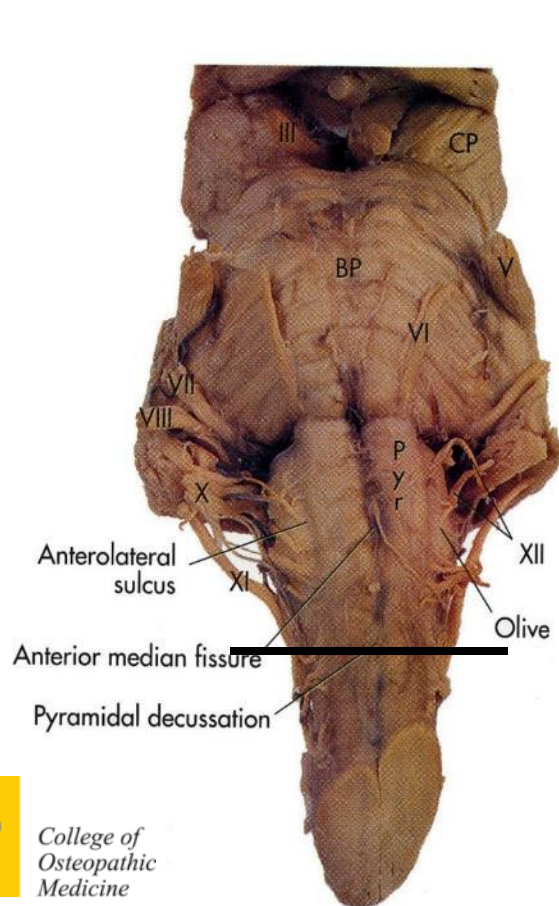
Spinothalamic



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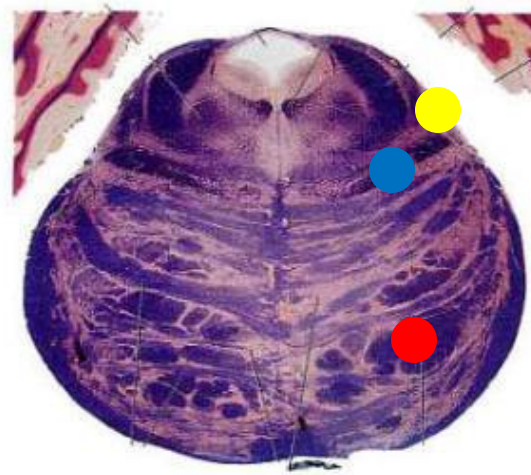




corticospinal

DCML

Spinothalamic



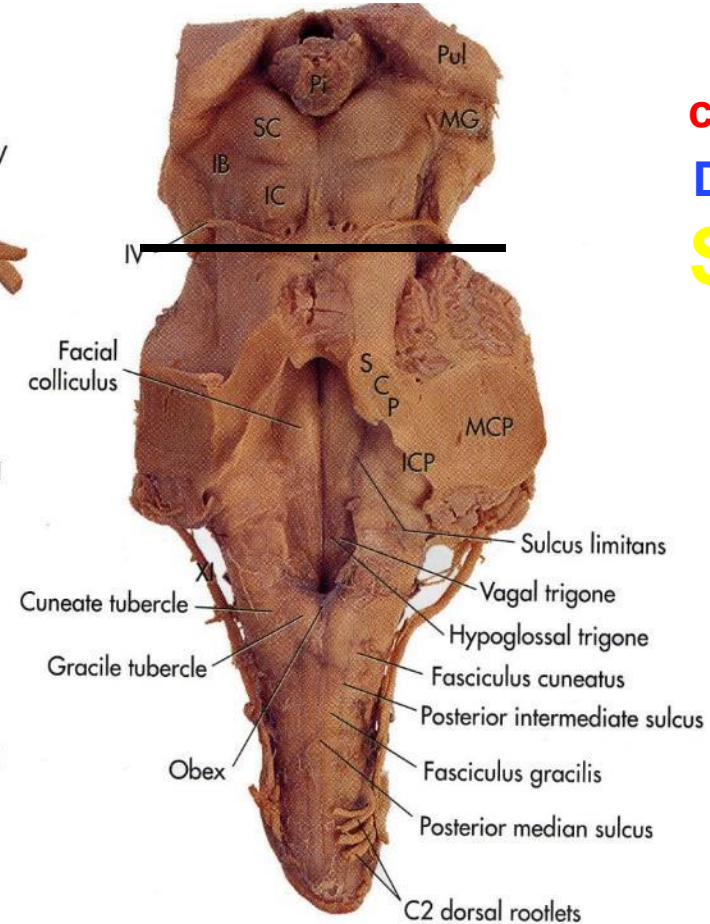
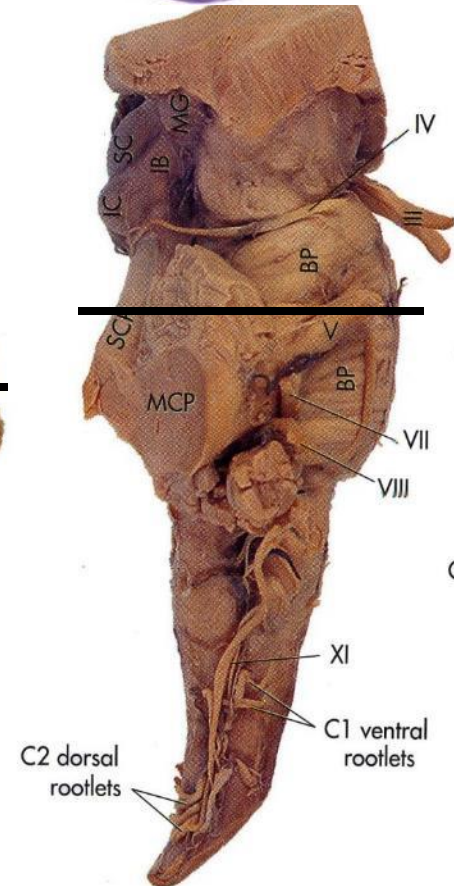
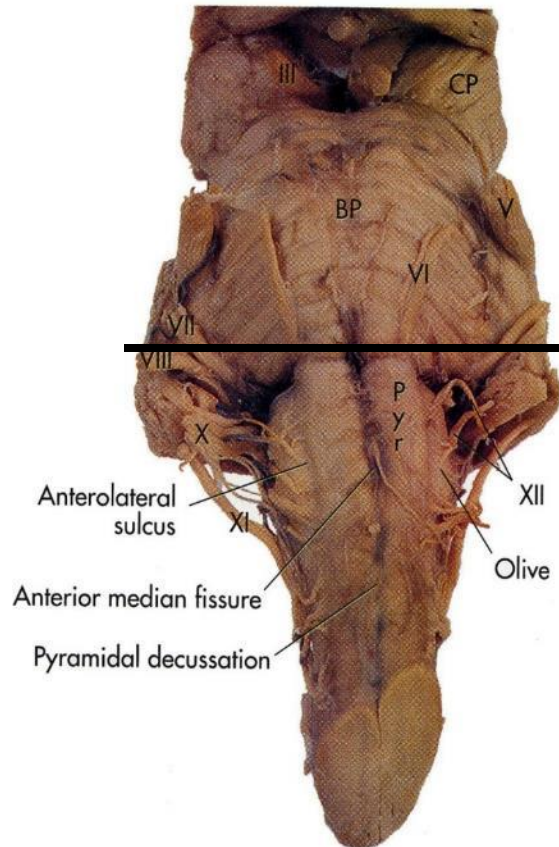
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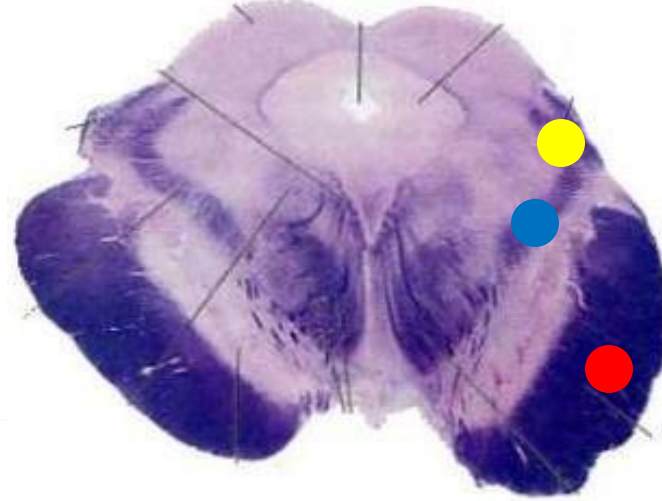
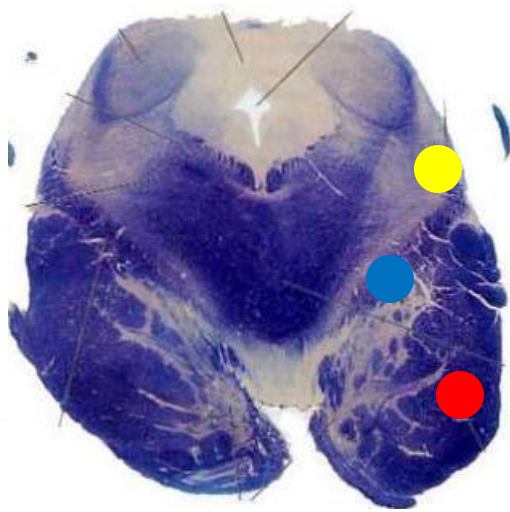
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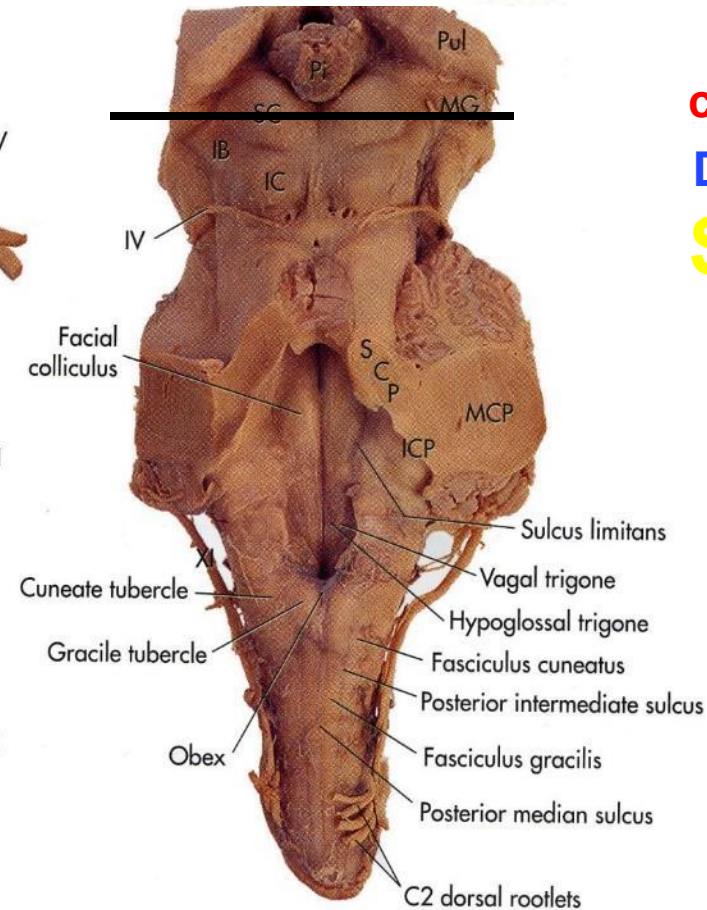
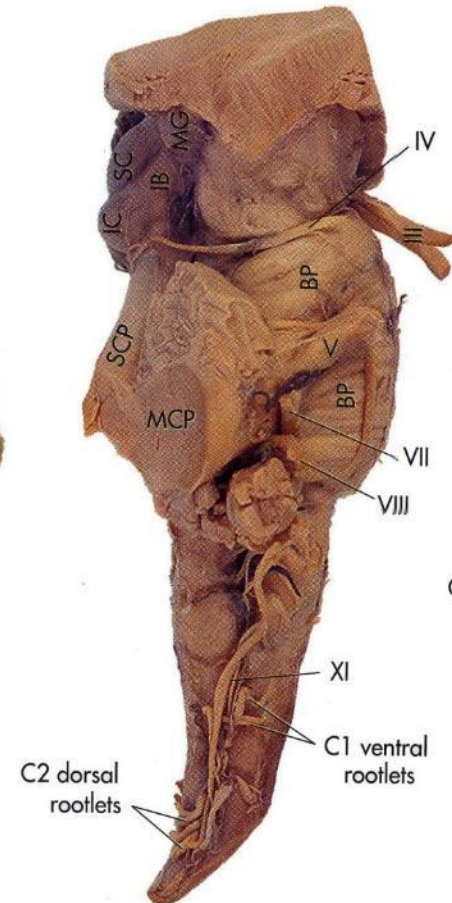
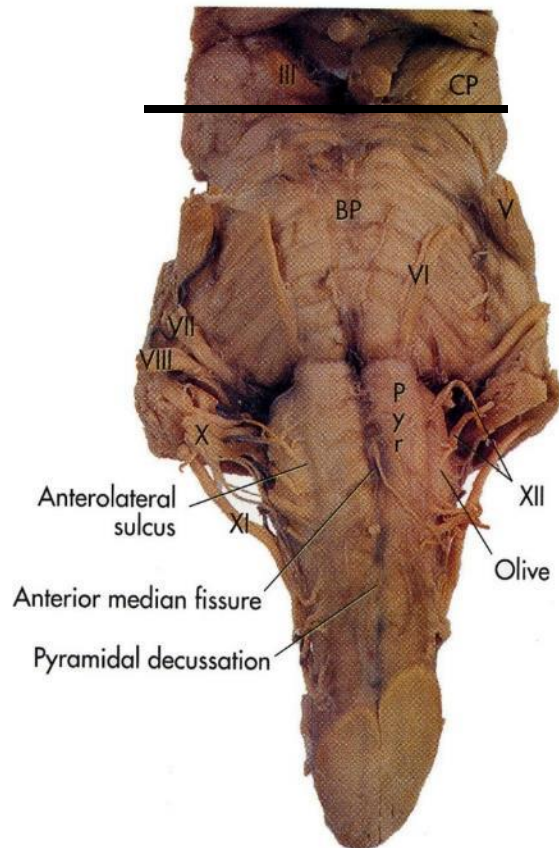
Spinothalamic





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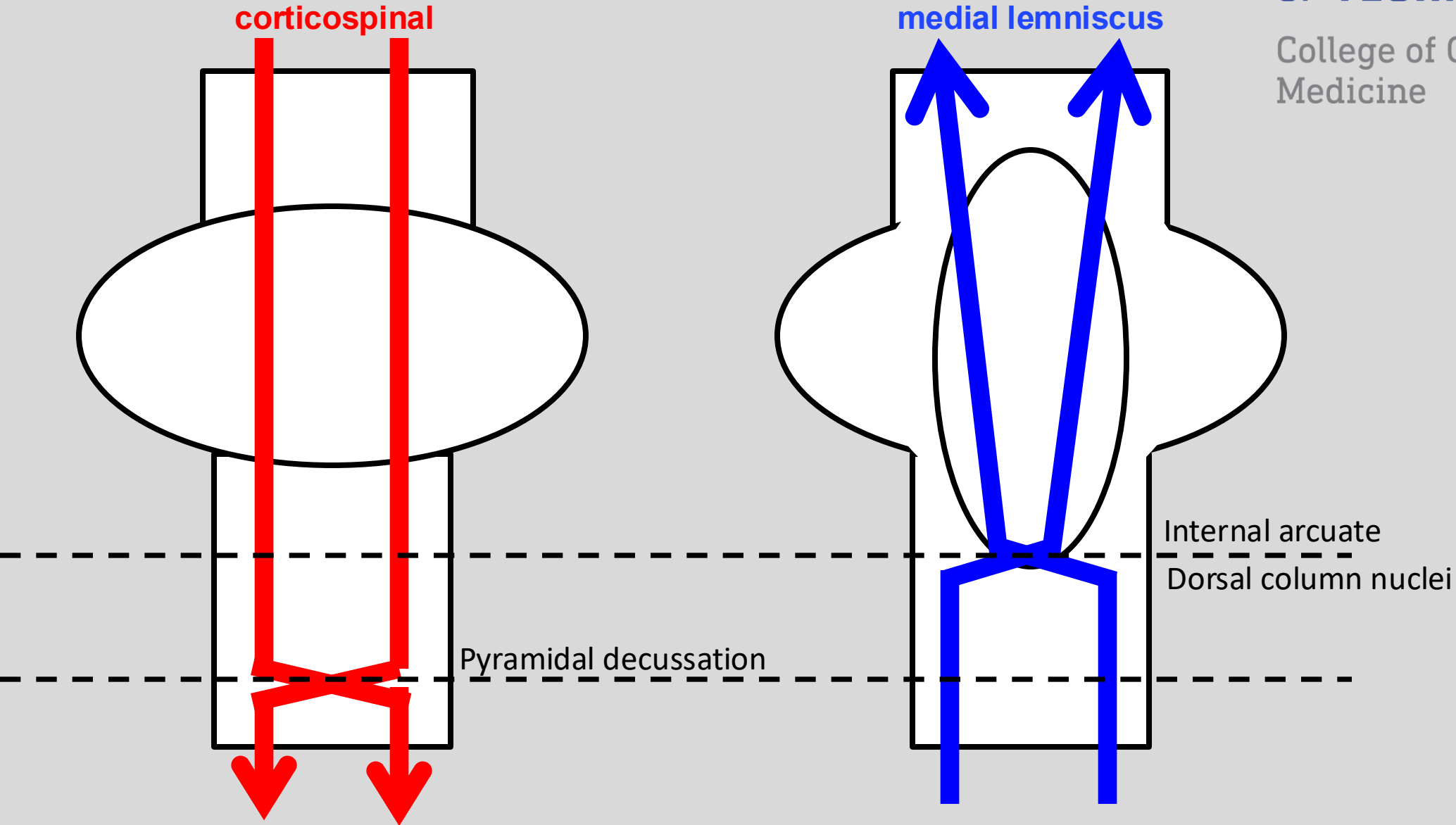
DCML

Spinothalamic

The Corticospinal tract decussates below the DCML decussation

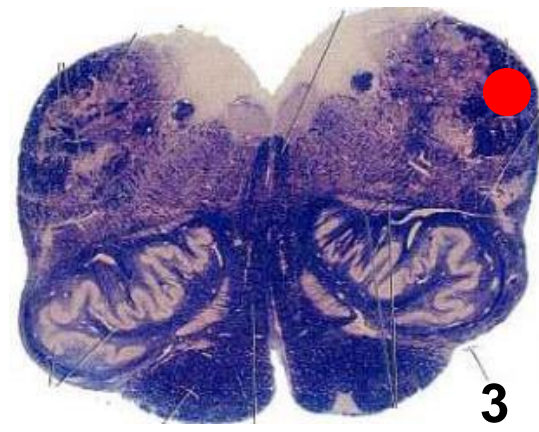
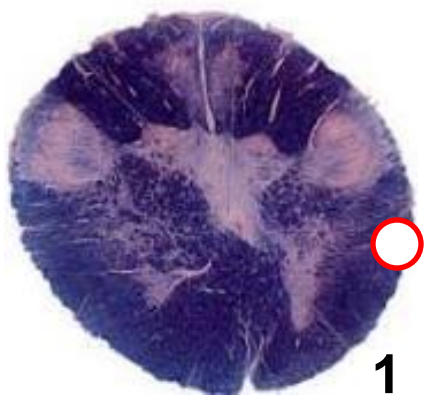
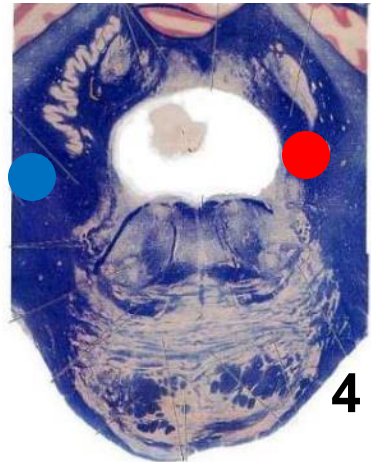
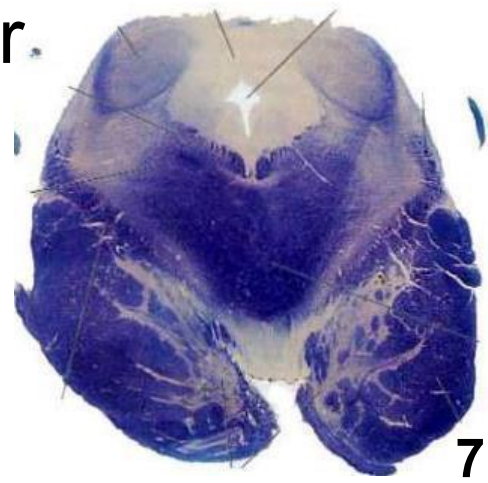
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Middle cerebellar.
Peduncle

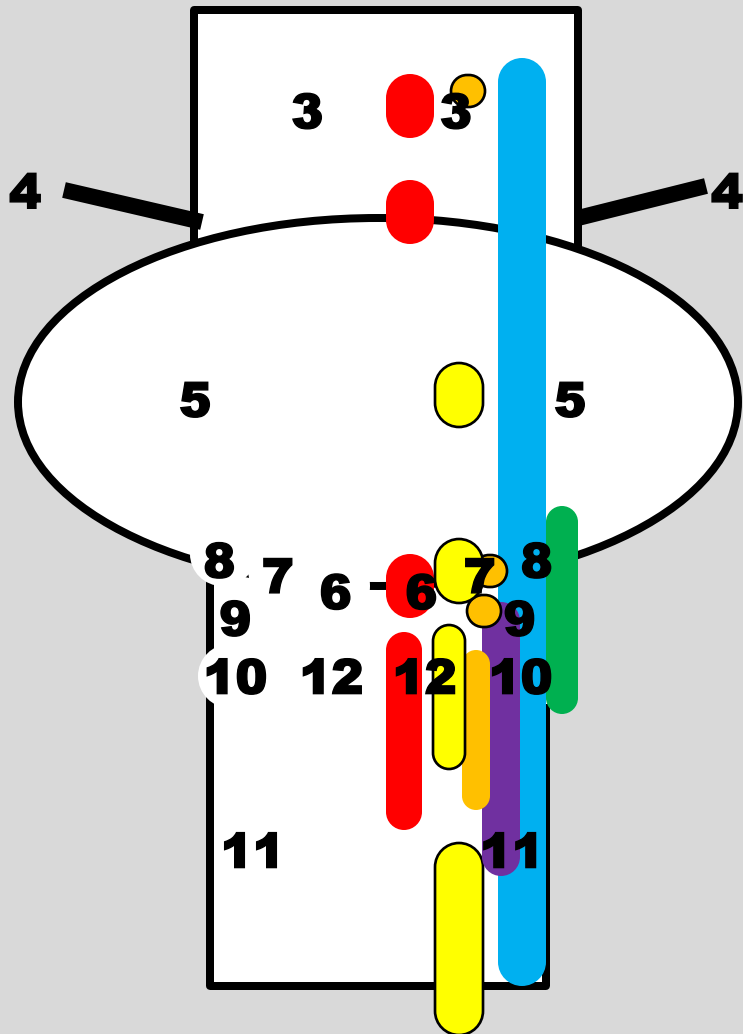
Inferior
Middle



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Cranial nerve nuclei have a consistent rostral-caudal position according to nerve and medial-lateral position according to function



somatic motor

brachial motor

visceral motor

visceral sensory

somatic sensory

special sensory

Motor medial

Sensory lateral

- Either motor or sensory, never both (unlike nerves!)
- Partially correspond to the same level as nerves

<i>cranial nerve nuclei summary</i>	<u>SOMATIC MOTOR</u>	<u>BRANCHIAL MOTOR</u>	<u>VISCERAL MOTOR</u>	<u>VISCERAL SENSORY</u>	<u>SOMATIC SENSORY</u>	<u>SPECIAL SENSORY</u>
III	oculomotor → eye muscles		Edinger Westphal → iris			
IV	trochlear → eye muscles					
V		trigeminal motor → jaw			jaw proprio → trigeminal mesencephalic face touch & proprio → trigeminal pontine face pain & temp → trigeminal spinal	
VI	abducens → eye muscles					
VII		facial motor → face & stapedius	superior salivatory → tears & saliva	ant. tongue taste → gustatory	posterior ext. ear → trigeminal spinal	
VIII						orientation → vestibular hearing → cochlear
IX		ambiguus → stylopharyngeus (swallow)	inferior salivatory → parotid gland (saliva)	posterior tongue taste → gustatory carotid baroreceptors → solitary	posterior tongue, tympanic mem → trigeminal spinal	
X		ambiguus → pharynx/larynx (swallow, gag reflex)	dorsal motor vagus → “rest and digest” autonomics	viscerosensory & aortic baroreceptors → solitary	ext. and int. ear → trigeminal spinal	
XI		spinal accessory → scm and upper trapezius				
XII	hypoglossal → tongue					

midbrain	Edinger Westphal	III	pupil constriction	visceral efferent
	oculomotor	III	eyes	somatic motor
	trig. mesencephalic	V	jaw	somatic sensory
	trochlear	IV	eyes	somatic motor
	trigeminal motor	V	jaw, upper neck	branchial motor
	trigeminal pontine	V	face	somatic sensory
	abducens	VI	eyes	somatic motor
	facial motor	VII	face, upper neck	branchial motor
	superior salivary	VII	mouth	visceral efferent
	vestibular	VIII	head motion / orientation	special sensory
pons	cochlear	VIII	hearing	special sensory
	inferior salivary	IX	parotid gland	visceral efferent
	solitary gustatory	{ VII	taste	visceral sensory
		{ IX, X	taste	visceral sensory
	solitary	{ IX	blood pressure, [O2]	sensory
		{ X	respiratory, cardio, digestive	visceral sensory
	hypoglossal	XII	tongue	somatic motor
	ambiguus*	{ IX, XI	swallowing	branchial motor
		{ X	throat, vocals	branchial motor
	spinal accessory**	XI	neck, shoulders	branchial motor
medulla	dorsal vagus	X	gastrointestinal, heart	visceral efferent
		{ V	face, oral cavity	somatic sensory
	trigeminal spinal	{ VII	ear	somatic sensory
		{ IX	throat, vocals	somatic sensory
		{ X	ear, pharynx	somatic sensory

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* ambiguus has a brachial component and a visceral efferent component to inhibit heart rate

**spinal accessory is sometimes classified as brachial and sometimes as somatic motor.

Summary

- The major descending and ascending tracts have distinct trajectories through the brainstem
- The corticospinal tract is always most ventral and decussates in the lower medulla
- The DCML decussates in the upper medulla and changes position through the brainstem but always more medial to the spinothalamic tract.
- The spinothalamic tract decussates in the spinal cord and is always lateral to the DCML.
- The three cerebellar peduncles are present at different levels of the brainstem; inferior – medulla and lower pons; middle – pons; superior – upper pons and lower midbrain.
- The cranial nerve nuclei have a rostral-caudal placement roughly aligned with the cranial nerves.
- Rational arrangement of cranial nerve nuclei: three motor columns are medial and three sensory columns are lateral.